Project Work – Elevator

Design an elevator control system using the Arduino MEGA (ATmega2560) as a master and Arduino UNO (ATmega328p) as a slave. The Arduino MEGA and Arduino UNO must communicate using suitable protocol, either SPI or I2C/TWI. **Submission created using Arduino IDE will not be graded.**

Project details:

- **Keypad control:** The elevator is controlled by a keypad to select floors 0–99
- **State machine:** The code must implement at least the following states:
  IDLE, ***GOING UP***, ***GOING DOWN***, ***DOOR OPENING***, ***DOOR CLOSING***, ***OBSTACLE DETECTION***, ***FAULT***
- **LCD feedback:** The current floor and relevant status are shown on the LCD
- **LED indicators:** LEDs indicate movement, door opening and closing, and obstacle detection
- **Buzzer:** When an obstacle is detected, the buzzer plays a melody

Suggested split: Arduino Mega controls the keypad and LCD. Arduino Uno controls the buzzer and LEDs. Other components may be controlled by either board as you prefer.

System functionality (state-by-state):

1. ***IDLE*:**
   a. Elevator waits for floor input from the keypad
   b. LCD displays "**Choose the floor**"
   c. After the floor is inputted, proceed to ***GOING UP*** or ***GOING DOWN***
2. ***GOING UP / GOING DOWN*:**
   a. Elevator moves to the inputted floor
   b. Movement LED is on until the floor is reached
   c. LCD screen displays "**Current floor: XX**" while moving
   d. On arrival to the inputted floor, proceed to ***DOOR OPENING***
3. ***DOOR OPENING*:**
   a. Door opening LED is on for 3 seconds
   b. LCD displays "**Door open**"
   c. After opening, either proceed to ***DOOR CLOSING*** or handle ***OBSTACLE DETECTION*** if triggered
4. ***OBSTACLE DETECTION*:**
   a. Trigger obstacle detection when the door is open. This can be via a sensor, an external button, or a special keypad character.
   b. Obstacle LED blinks three times
   c. LCD displays "**Obstacle detected**"
   d. The buzzer plays a melody that consist at least 5 notes. The melody will stop only if any button on the keypad is pressed.
   e. System goes back to "***DOOR CLOSING***"

5. ***DOOR CLOSING***
   a. Door closing LED is on for 2 seconds
   b. LCD displays "**Door closing**"
   c. Return to ***IDLE***
6. ***FAULT***
   a. If selected floor equals the current floor, enter ***FAULT*** state
   b. LCD displays "**Same floor**"
   c. System goes to ***IDLE***

Additional functionalities for the project work (max 5 points):

Queued floor requests: Store multiple floor requests and process them in order	**2 points**
Energy saving mode: System can enter sleep state with < 1 mA current, and wake up without rebooting	**1 point**
Non-stop buzzer: Jingle that runs constantly without disrupting core logic	**1 point**
Add your own functionalities	**1 point**

Library and software rules:

- You may use the external libraries or header files for LCD and keypad that are provided on Moodle, and modify them if wanted
- You may use libraries that ship with Microchip Studio, such as <util/delay>
- **Submissions created with Arduino IDE will not be graded**

Project work is divided into two mandatory sections: **Demo session** and **Report (code submission)**.

Demo session

- **Format:** in-person only in Lappeenranta or Lahti
- **Attendance:** personal attendance of **each student** is required to receive points
  - Missing the demo results in a **personal deduction of 50%** of the group's points
- **What to demonstrate**
  - The circuit's functionalities
  - Each member explains their individual contribution and the corresponding code parts, consistent with the roles described in the report
    - If a member demonstrates insufficient understanding, group's points of the section will be deducted
- **Scheduling**
  - The demo schedule will be announced later. To present outside the scheduled time, contact the course assistants to arrange a suitable slot

Report and code

- **What to submit**
  - The **entire commented Microchip Studio project**, including main.c and any self-made or modified .c or .h files
  - A brief report of **about 5 pages** in **PDF** format
- **Report contents:**
  - **Contributions:** a clear explanation of the contribution of each student. In the demo, each student presents their own section
  - **Circuit diagram:** include pin connections
  - **Functionality summary:** a concise explanation of how the system works
- **Commenting requirement**
  - Code should be thoroughly commented before the demo. But if needed, code comments may still be improved **until the final submission deadline**
- **File naming:**
  - Group_NN.zip

Grading:

To receive a grade from the course, the group must return their Arduino kits in a proper arrangement. This can be done directly after the demo. Any broken or missing components should be reported. Read the "Project work grading" from Moodle for the evaluation criteria.

DEMO DEADLINE IS: 8.5.2026 AT 15.00

REPORT DEADLINE IS: 10.5.2026 AT 23:59

Every late submission will be deducted 2 point/day